KYZEN AQUANOX A4639

Advanced Aqueous Cleaning Agent — Technical Guide

Water-Based Cleaning Chemistry • PCBA Flux Removal • Lead-Free & Leaded Compatible

Process Parameters • Bath Maintenance • Environmental Compliance

PCBA In-House Cleaning Chemistry & Process Standards

IPC/J-STD-001 | IPC-A-610 | RoHS | REACH

1. KYZEN AQUANOX A4639 OVERVIEW

KYZEN® AQUANOX® A4639 is a **high-performance, water-based (aqueous) cleaning agent** specifically formulated for the electronics assembly industry. It is designed to remove **water-soluble (WS) and no-clean (NC) flux residues** from printed circuit board assemblies (PCBA) after reflow or wave soldering processes.

Product Classification:

- **Type:** Aqueous-based alkaline cleaning chemistry
- **pH (concentrate):** ~11.0–12.5 (alkaline)
- **Dilution:** Typically 10–20% in DI water (verify KYZEN process recommendation)
- **Flash Point:** None (water-based, non-flammable)
- **RoHS Status:** Compliant — contains no restricted substances
- **REACH Status:** Compliant
- **Biodegradability:** High — environmentally responsible formulation

Key Advantages:

- **Excellent flux removal:** Effective on water-soluble and no-clean residues
- **Low foaming:** Suitable for spray-in-air and ultrasonic cleaning equipment
- **Material safe:** Compatible with aluminum, copper, tin, and most plastics
- **Easy to rinse:** Water-soluble — rinses completely with DI water
- **Long bath life:** Stable chemistry with consistent cleaning performance
- **Low VOC:** Minimal volatile organic compound emissions
- **Non-flammable:** Safe for factory floor use without special fire precautions

■ **In-House Positioning:** AQUANOX A4639 complements the DI water cleaning system described in the previous guide. While DI water alone removes water-soluble fluxes, A4639 adds chemical cleaning power for stubborn no-clean residues and heavy contamination that DI water cannot fully remove.

2. CHEMISTRY & CLEANING MECHANISM

2.1 Formulation

AQUANOX A4639 is a **proprietary blend** of surfactants, saponifiers, and alkaline builders dissolved in water. The exact formulation is KYZEN intellectual property, but the cleaning mechanism is well understood:

Key Components:

- **Alkaline builders:** Raise pH to saponify (convert to soap) rosin and resin flux residues
- **Surfactants:** Reduce surface tension to penetrate under components and wet surfaces
- **Saponifiers:** Chemically react with organic acids in flux to form water-soluble soaps
- **Water:** Carrier and rinse medium
- **Corrosion inhibitors:** Protect aluminum, copper, and other sensitive metals

2.2 Cleaning Mechanism

The cleaning action of AQUANOX A4639 occurs through three simultaneous mechanisms:

1. Chemical Saponification

Alkaline agents react with rosin/resin acids in flux residues to form water-soluble soaps. This converts insoluble organic contamination into soluble compounds that can be rinsed away.

2. Surfactant Action

Surfactants reduce the surface tension of the cleaning solution, allowing it to penetrate under low-standoff components (QFN, BGA, CSP) and into tight spaces where mechanical spray cannot reach.

3. Mechanical Agitation

Spray pressure or ultrasonic cavitation provides mechanical energy to dislodge particles and transport dissolved contaminants away from the board surface.

2.3 pH and Concentration Relationship

Concentration	pH Range	Cleaning Strength	Recommended For
5%	~10.5–11.0	Light	Light no-clean residues, maintenance cleaning
10%	~11.0–11.5	Moderate	Standard no-clean fluxes, general PCBA
15%	~11.5–12.0	Strong	Heavy no-clean, water-soluble fluxes
20%	~12.0–12.5	Maximum	Stubborn residues, rework, heavy contamination

Table 1: AQUANOX A4639 Concentration vs. pH and Cleaning Strength

3. PROCESS PARAMETERS

3.1 Concentration

The optimal concentration depends on the flux type, contamination level, and cleaning equipment. Typical in-house starting point is **10–15%** in DI water.

- **Water-soluble fluxes (WS-820, WS-809, WS715M):** 10–15%
- **No-clean fluxes (OM-353, RF-800):** 15–20%
- **RMA fluxes:** 15–20%
- **Heavy contamination / rework:** 20%

3.2 Temperature

Higher temperatures improve cleaning kinetics and reduce surface tension, but excessive heat can damage sensitive components or accelerate bath degradation.

- **Wash Temperature:** 60–70°C (140–158°F)
- **Optimal:** 65°C for most applications
- **Maximum:** 75°C (risk of component damage and excessive foaming)
- **Rinse Temperature:** 60–70°C (warm rinse improves drying)
- **Final Rinse:** Room temperature to 40°C (prevents water spots)

3.3 Wash Time

Wash time depends on flux type, board density, and contamination level.

- **Light contamination:** 3–5 minutes
- **Standard no-clean:** 5–10 minutes
- **Heavy no-clean / RMA:** 10–15 minutes
- **Rework / repair:** 15–20 minutes

3.4 Spray Pressure

Mechanical energy from spray pressure assists chemical cleaning action.

- **Pre-wash:** 1–2 bar
- **Main wash:** 2–4 bar
- **Rinse:** 1–3 bar
- **Final rinse:** 0.5–1 bar

3.5 Rinse Requirements

Thorough rinsing is critical — AQUANOX A4639 residues are alkaline and must be completely removed to prevent long-term corrosion or electrical issues.

- **Rinse stages:** Minimum 2 stages (wash water + fresh DI water)
- **DI water quality:** ≥ 1 MΩ·cm (18 MΩ·cm for final rinse)
- **Rinse time:** 2–5 minutes per stage

• **Final rinse pH check:** Target pH 6.5–7.5 (neutral)

■■ **Critical:** Incomplete rinsing leaves alkaline residues that can cause corrosion, electrochemical migration, and conformal coating adhesion failure. Always verify final rinse pH and conductivity before drying.

4. EQUIPMENT COMPATIBILITY

4.1 Spray-in-Air Cleaning Systems

AQUANOX A4639 is **low-foaming** and well-suited for spray-in-air batch and inline cleaning systems. The surfactant package is designed to break down foam under high-pressure spray conditions.

Compatible Equipment Types:

- **Batch spray systems:** Excellent — low foam, good drainage
- **Inline conveyor washers:** Excellent — stable at 60–70°C
- **Rotary drum washers:** Good — verify foam control at high agitation
- **High-pressure spray:** Good up to 5 bar; test above 5 bar for foam

4.2 Ultrasonic Cleaning Systems

AQUANOX A4639 can be used in ultrasonic baths, but the combination of ultrasonics + alkaline chemistry requires caution for sensitive components.

- **Ultrasonic frequency:** 40 kHz standard; 28 kHz for heavy contamination
- **Power density:** 10–20 W/L (lower than solvent cleaning to prevent cavitation damage)
- **Component caution:** Avoid ultrasonics on MEMS, crystal oscillators, and fine-wire bonded devices
- **Bath life:** Monitor pH and contamination load; replace when pH drops > 0.5 units

4.3 Material Compatibility

Material	Compatibility	Precautions
Aluminum	■ Good	Corrosion inhibitors protect; avoid prolonged immersion > 30 min
Copper / ENIG	■ Excellent	No issues at recommended concentration and temperature
Tin / SAC305	■ Excellent	No galvanic corrosion at pH < 12.5
Silver	■■ Caution	Tarnishing possible at high concentration / temperature / time
Nickel	■ Good	Generally safe; monitor for discoloration
FR-4 PCB	■ Excellent	No delamination or swelling at < 75°C
Polyimide (Kapton)	■ Good	No degradation at process parameters
Acrylic Conformal Coating	■ Incompatible	Alkaline chemistry attacks acrylic; remove coating before cleaning

Silicone Conformal Coating	■ Good	Resistant to alkaline cleaning
Polycarbonate	■■ Caution	Stress cracking possible at high pH and temperature
ABS Plastic	■■ Caution	Monitor for surface degradation

Table 2: AQUANOX A4639 Material Compatibility Guide

5. BATH MAINTENANCE & MONITORING

5.1 Key Monitoring Parameters

Regular monitoring ensures consistent cleaning performance and prevents board damage from degraded chemistry.

Parameter	Frequency	Target	Action if Out of Spec
Concentration	Every 4 hours	10–20%	Add concentrate or fresh solution
pH	Every 4 hours	11.0–12.5	Replace bath if pH < 10.5
Conductivity	Every 4 hours	Baseline + 20%	Replace when > 150% of baseline
Temperature	Continuous	60–70°C	Adjust heater; check thermostat
Contamination Load	Daily	< 5% flux loading	Replace bath or filter
Foam Level	Every batch	Minimal	Add anti-foam or reduce agitation

Table 3: AQUANOX A4639 Bath Monitoring Schedule

5.2 Concentration Measurement Methods

Accurate concentration measurement is essential for process control.

Method 1: Refractive Index (Recommended)

- Use handheld refractometer (0–10% or 0–20% scale)
- Calibrate with known concentration standards
- Quick, accurate, no chemical handling

Method 2: Titration

- Acid-base titration with standardized HCl
- More accurate than refractometer for aged baths
- Requires lab equipment and trained operator

Method 3: Conductivity

- Correlate conductivity to concentration for your water source
- Quick check but less accurate due to contamination interference
- Best used as a trend indicator, not absolute measurement

5.3 Bath Replacement Criteria

Replace the AQUANOX A4639 bath when ANY of the following occur:

- **pH drops below 10.5** (cleaning effectiveness compromised)
- **Conductivity exceeds 150%** of baseline (excessive contamination)

- **Cleaning effectiveness degrades** (visible residues after cleaning)
- **ROSE test fails** on cleaned boards (ionic contamination too high)
- **Visible oil layer or sludge** forms on bath surface
- **Odor changes significantly** (bacterial or chemical degradation)

■ **Best Practice:** Maintain a bath log tracking concentration, pH, conductivity, temperature, number of boards cleaned, and replacement dates. This data enables predictive maintenance and demonstrates process control to customers and auditors.

6. QUALITY VERIFICATION AFTER CLEANING

6.1 Visual Inspection

First-line quality check immediately after cleaning and drying.

- **Acceptable:** No visible white residues, no water spots, no discoloration
- **Acceptable:** No component displacement or damage
- **Reject:** White powdery residue (flux residue or cleaner residue)
- **Reject:** Water spots or streaks (incomplete drying or poor rinse)
- **Reject:** Board warpage or delamination

6.2 ROSE Testing (Resistivity of Solvent Extract)

The industry-standard method for quantifying ionic cleanliness.

- **IPC-TM-650 2.3.25:** Standard test method
- **Class 2 (Dedicated Service):** ≤ 1.56 µg/cm² NaCl equivalent
- **Class 3 (High Performance):** ≤ 0.78 µg/cm² NaCl equivalent
- **Frequency:** 1 board per lot for Class 2; 100% for Class 3

6.3 Ion Chromatography (IC)

For root-cause analysis when ROSE fails or specific ion identification is needed.

- **Detects:** F■, Cl■, Br■, NO■■, PO■³■, SO■²■, Na■, K■, NH■■
- **Detection limit:** 0.01 ppm
- **Use case:** Identify source of contamination (flux, cleaner, water, handling)

6.4 Surface Insulation Resistance (SIR)

Long-term reliability test for high-performance applications.

- **Test conditions:** 85°C / 85% RH for 7 days
- **Minimum SIR:** ≥ 10■ Ω (100 MΩ)
- **Use case:** Automotive (Tesla, Zoox), medical, military

■■ **Critical:** If ROSE or SIR fails after AQUANOX A4639 cleaning, check: (1) rinse completeness — verify final rinse pH is neutral, (2) DI water quality — confirm ≥ 1 MΩ·cm, (3) bath concentration — may be too low or contaminated, (4) dry time — boards must be completely dry before testing.

7. TROUBLESHOOTING GUIDE

7.1 White Residue After Cleaning

Cause 1: Insufficient rinse — alkaline cleaner residue left on board

→ **Solution:** Extend rinse time; add rinse stage; verify DI water quality

Cause 2: AQUANOX concentration too high

→ **Solution:** Reduce to 10–15%; verify with refractometer

Cause 3: Water evaporating before complete rinse

→ **Solution:** Increase rinse spray pressure; reduce conveyor speed

7.2 Incomplete Flux Removal

Cause 1: AQUANOX concentration too low

→ **Solution:** Increase to 15–20%; verify with refractometer

Cause 2: Wash temperature too low

→ **Solution:** Raise to 65–70°C; verify heater operation

Cause 3: Wash time too short

→ **Solution:** Extend to 10–15 minutes for no-clean fluxes

Cause 4: Bath contaminated / depleted

→ **Solution:** Replace bath; check pH and conductivity

Cause 5: Flux type incompatible with aqueous cleaning

→ **Solution:** Some no-clean fluxes require semi-aqueous or solvent cleaning

7.3 Excessive Foaming

Cause 1: Spray pressure too high

→ **Solution:** Reduce to 2–3 bar; increase temperature to reduce foam

Cause 2: Bath contaminated with oils or surfactants

→ **Solution:** Replace bath; check for upstream contamination

Cause 3: Water hardness too high

→ **Solution:** Verify DI water system; check softener operation

7.4 Board Discoloration

Cause 1: Temperature too high or time too long

→ **Solution:** Reduce to 60–65°C; reduce wash time to 5–8 minutes

Cause 2: Incompatible material (e.g., silver, certain plastics)

→ **Solution:** Verify material compatibility; reduce concentration to 5–10%

Cause 3: Oxidation of exposed copper

→ **Solution:** Add corrosion inhibitor or reduce temperature; ensure rapid drying

8. ENVIRONMENTAL & SAFETY

8.1 Environmental Profile

AQUANOX A4639 is designed for environmental responsibility:

- **VOC Content:** < 50 g/L (very low compared to solvent cleaners)
- **Biodegradability:** > 90% in 28 days (OECD 301 test)
- **Aquatic Toxicity:** Low — safe for wastewater treatment systems
- **Contains:** No halogenated solvents, no aromatic hydrocarbons
- **GHS Classification:** Irritant (skin and eye); not classified as corrosive

8.2 Waste Water Disposal

Spent AQUANOX A4639 solution and rinse water may require treatment before discharge:

- **pH adjustment:** Neutralize to pH 6–9 before discharge
- **Heavy metals:** Test for lead, tin, silver; treat if above local limits
- **COD (Chemical Oxygen Demand):** May be elevated due to dissolved flux
- **Evaporation:** Reduce volume via evaporator; dispose concentrate as hazardous
- **Contract disposal:** Use certified waste management for concentrated waste

8.3 Operator Safety

AQUANOX A4639 is alkaline and requires standard chemical handling precautions:

- **Skin contact:** Wear nitrile or neoprene gloves
- **Eye protection:** Safety glasses or face shield during bath maintenance
- **Inhalation:** Low vapor pressure — minimal inhalation risk at process temperature
- **Spills:** Contain with absorbent; neutralize with dilute acid if large spill
- **First aid:** Skin — wash with water; Eyes — flush 15 minutes, seek medical attention

■■ **Safety Note:** AQUANOX A4639 is alkaline (pH 11–12.5). Prolonged skin contact can cause irritation. Always wear gloves when handling concentrate or working in cleaning equipment. Eye contact requires immediate flushing and medical evaluation.

9. IN-HOUSE PROCESS INTEGRATION

9.1 Integration with Existing DI Water System

AQUANOX A4639 integrates with the existing in-house DI water cleaning infrastructure:

- **DI water source:** Same DI generation system (≥ 1 MΩ·cm)
- **Cleaning equipment:** Batch ultrasonic or inline spray washers
- **Rinse system:** Existing DI water rinse tanks and spray headers
- **Dry system:** Existing hot air or IR dryers
- **Quality testing:** Same ROSE, SIR, and visual inspection protocols

9.2 Recommended Process Flow

1. **Reflow / Wave Solder** — Complete soldering first
2. **Cool Down** — Allow boards to cool to < 60°C
3. **Pre-wash** — DI water rinse to remove loose flux (optional)
4. **AQUANOX Wash** — 10–15% solution, 65°C, 5–10 minutes
5. **DI Water Rinse 1** — Remove bulk cleaner
6. **DI Water Rinse 2** — High-purity rinse (18 MΩ·cm)
7. **Final Rinse pH Check** — Verify neutral pH 6.5–7.5
8. **Dry** — Hot air 80–120°C, 10–30 minutes
9. **Visual Inspection** — Check for residues and damage
10. **ROSE Test** — Verify ionic cleanliness
11. **Next Process** — Conformal coating, ICT, or packaging

9.3 Flux-Specific Recommendations

In-House Flux / Paste	AQUANOX Conc.	Temp	Time	Rinse Stages
ALPHA WS-820 (Water-soluble SAC305)	10–15%	65°C	5–8 min	2
ALPHA WS-809 (Water-soluble SnPb)	10–15%	65°C	5–8 min	2
ALPHA OM-353 (No-clean SAC305)	15–20%	65–70°C	10–15 min	3
ALPHA RF-800 (No-clean flux)	15–20%	65–70°C	10–15 min	3
AIM WS715M (Water-soluble flux)	10–15%	65°C	5–8 min	2
RMA / Rosin Flux	15–20%	70°C	10–15 min	3

Table 4: AQUANOX A4639 Process Recommendations by In-House Flux Type

■ **In-House Best Practice:** For no-clean fluxes (OM-353, RF-800) that customer requires to be cleaned, use AQUANOX A4639 at 15–20% with 3 rinse stages. No-clean residues are more difficult to remove than water-soluble fluxes and require stronger chemistry and longer wash times.

10. QUICK REFERENCE SUMMARY

Product:

→ **KYZEN AQUANOX A4639** — Aqueous alkaline cleaning agent for PCBA flux removal

Chemistry:

→ **Type:** Water-based, alkaline (pH 11.0–12.5 concentrate)

→ **Mechanism:** Saponification + surfactant penetration + mechanical agitation

→ **RoHS/REACH:** Fully compliant

Process Parameters:

→ **Concentration:** 10–20% in DI water (10–15% WS; 15–20% no-clean)

→ **Temperature:** 60–70°C (optimal 65°C)

→ **Wash time:** 5–15 minutes (flux-dependent)

→ **Spray pressure:** 2–4 bar (main wash)

→ **Rinse:** Minimum 2 stages; final rinse 18 MΩ·cm DI water

→ **Dry:** 80–120°C hot air, 10–30 minutes

Bath Maintenance:

→ **Monitor:** Concentration, pH, conductivity, temperature every 4 hours

→ **Replace when:** pH < 10.5, conductivity > 150% baseline, or cleaning degrades

→ **Measurement:** Refractometer (quick), titration (accurate), conductivity (trend)

Quality Verification:

→ **Visual:** No white residue, no water spots, no damage

→ **ROSE:** ≤ 1.56 $\mu g/cm^2$ (Class 2); ≤ 0.78 $\mu g/cm^2$ (Class 3)

→ **SIR:** ≥ 10■ Ω at 85°C/85% RH for 7 days

→ **Final rinse pH:** 6.5–7.5 (neutral)

Safety:

→ **Gloves:** Nitrile or neoprene mandatory

→ **Eyes:** Safety glasses during bath maintenance

→ **Spills:** Contain and neutralize; flush skin/eyes immediately if contact

→ **Waste:** Neutralize pH to 6–9 before discharge; test for heavy metals

References & Standards:

• KYZEN Technical Datasheet: AQUANOX A4639

• IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies

- IPC-A-610: Acceptability of Electronic Assemblies
- IPC-TM-650 2.3.25: Resistivity of Solvent Extract (ROSE) Test
- IPC-TM-650 2.6.3.3: Surface Insulation Resistance (SIR) Test
- MIL-STD-202 Method 310: Contamination Testing
- GHS Classification: Globally Harmonized System of Classification and Labeling of Chemicals
- REACH Regulation (EC) No 1907/2006
- RoHS Directive 2011/65/EU

End of Document — KYZEN AQUANOX A4639 Complete Technical Guide